

Forecasting Project Final Report: U.S. CPI Inflation

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```
library(fredr)
library(tidyverse)
library(forecast)
library(tseries)
library(knitr)

fredr_set_key("b09b5e2628663eedfd73728ebbb49cb0")
```

1. Project Overview

This report evaluates the out-of-sample forecast submitted in the Data Report against the actual March 2026 CPI release. The forecasting target was U.S. headline CPI inflation (CPIAUCSL, year-over-year percentage change), sourced from FRED. The model used was a log-differenced ARIMA with a COVID dummy regressor, selected via `auto.arima()` and validated with the Ljung-Box test.

The BLS released the March 2026 CPI reading on April 10, 2026. This report compares that result to the submitted forecast, calculates accuracy metrics, checks whether the actual fell within the prediction intervals, and discusses what the outcome reveals about the model's strengths and limitations.

2. Actual vs. Forecast

```
# Submitted forecasts from Data Report
headline_point <- 2.68
headline_lo80  <- 2.35
headline_hi80  <- 3.01
headline_lo50  <- 2.50
headline_hi50  <- 2.85

core_point     <- 2.72
core_lo80      <- 2.59
core_hi80      <- 2.85
core_lo50      <- 2.65
core_hi50      <- 2.78
```

```

# Actual March 2026 BLS release (April 10, 2026)
actual_headline <- 3.3
actual_core     <- 2.6

error_headline <- actual_headline - headline_point
error_core     <- actual_core - core_point

results <- data.frame(
  Series   = c("Headline CPI", "Core CPI"),
  Forecast = c(headline_point, core_point),
  Actual   = c(actual_headline, actual_core),
  Error    = round(c(error_headline, error_core), 2),
  In_50 = ifelse(c(
    actual_headline >= headline_lo50 & actual_headline <= headline_hi50,
    actual_core     >= core_lo50    & actual_core     <= core_hi50
  ), "Yes", "No"),
  In_80 = ifelse(c(
    actual_headline >= headline_lo80 & actual_headline <= headline_hi80,
    actual_core     >= core_lo80    & actual_core     <= core_hi80
  ), "Yes", "No")
)

kable(results,
  col.names = c("Series", "Point Forecast (%)", "Actual (%)",
    "Error (pp)", "In 50% Interval?", "In 80% Interval?"),
  align = "lrrrcc",
  caption = "Forecast vs. Actual: March 2026 CPI Release")

```

Table 1: Forecast vs. Actual: March 2026 CPI Release

Series	Point Forecast (%)	Actual (%)	Error (pp)	In 50% Interval?	In 80% Interval?
Headline CPI	2.68	3.3	0.62	No	No
Core CPI	2.72	2.6	-0.12	No	Yes

The headline model underforecast by 0.62 percentage points and the actual value fell just outside the 80% prediction interval of [2.35, 3.01]. The core model performed well, overforecasting by only 0.12 percentage points with the actual falling inside the 80% interval but just below the 50% interval lower bound of 2.65%. The gap in performance between headline and core is the central finding of this evaluation and is discussed in Section 4.

3. Accuracy Metrics

With only one out-of-sample observation, MAE and RMSE are equal to the absolute and squared forecast error respectively. Both are reported for completeness.

```

mae_headline <- abs(error_headline)
rmse_headline <- sqrt(error_headline^2)

mae_core <- abs(error_core)
rmse_core <- sqrt(error_core^2)

accuracy_table <- data.frame(
  Series = c("Headline CPI", "Core CPI"),
  MAE = round(c(mae_headline, mae_core), 4),
  RMSE = round(c(rmse_headline, rmse_core), 4)
)

kable(accuracy_table,
  col.names = c("Series", "MAE (pp)", "RMSE (pp)"),
  align = "lrr",
  caption = "Forecast Accuracy: March 2026")

```

Table 2: Forecast Accuracy: March 2026

Series	MAE (pp)	RMSE (pp)
Headline CPI	0.62	0.62
Core CPI	0.12	0.12

The headline MAE of 0.62 percentage points is a meaningful miss relative to a forecast of 2.68%, representing roughly a 23% proportional error. The core MAE of 0.12 percentage points is negligible. As discussed in Section 4, the difference in accuracy between the two series is entirely explained by the energy price shock, not by a flaw in the modeling approach.

4. Model Evaluation

4.1 ARIMA Performance and Model Selection

The model from the Data Report was a univariate ARIMA, meaning it used only the historical pattern of CPI inflation itself to generate forecasts. ARIMA is well suited for inflation because the series is persistent: this month's reading is strongly correlated with recent past months, so the model can capture that structure and project it forward.

Model selection used `auto.arima()` with `stepwise = FALSE` and `approximation = FALSE`, which performs a thorough search over candidate ARIMA orders and selects the specification with the lowest AIC. The selected headline model was ARIMA(0,0,3)(0,0,1)[12], which includes a seasonal moving average term at lag 12. This seasonal component was retained by `auto.arima()` because monthly CPI data exhibits recurring seasonal patterns, and the [12] term directly captures that annual cycle. A naive random walk benchmark would produce a flat forecast with no seasonal adjustment; the seasonal ARIMA outperforms this by accounting for both the autocorrelation structure and the seasonal pattern in a single model.

The ADF test confirmed the correct integration order before modeling (Headline YoY: $p = 0.042$; Log CPI: $p = 0.947$ at level, $p = 0.01$ at first difference), and residual diagnostics confirmed the model fully absorbed the available autocorrelation in the data (Ljung-Box $p = 0.1749$ for headline, $p = 0.1531$ for core).

The core CPI forecast came in at 2.72% against an actual of 2.60%, a miss of just 0.12 percentage points. Core CPI strips out food and energy prices, so it moves more slowly and predictably than headline. With no energy shock distorting the signal, the ARIMA model captured the underlying inflation trend accurately.

4.2 Spurious Regression and Stationarity

Before fitting any model, it is essential to verify whether the series is stationary. Regressing non-stationary $I(1)$ series without testing first produces what is known as spurious regression, where the model appears to find a significant relationship even when none actually exists. Simulation studies show this false positive rate reaching about 75% with non-stationary data, compared to the expected 5%. The consequence is that t-statistics and R-squared values become unreliable guides to model quality.

The Data Report addressed this by running ADF tests at the level, first difference, and second difference for both series. The results confirmed that modeling the first difference of log CPI was the correct specification. The COVID dummy variable served a related purpose: without it, the extreme 2021-2022 inflation episode introduced residual autocorrelation that caused the Ljung-Box test to fail, signaling the model had not fully absorbed the structure in the data.

4.3 ARIMAX as a Potential Extension

The headline underforecast is fundamentally an omitted variable problem. Gas prices, which surged due to the Iran conflict, are a leading indicator for headline CPI that a univariate ARIMA cannot account for. ARIMAX addresses this by adding an external regressor to the ARIMA framework via the `xreg` argument, while `auto.arima()` still selects the best ARIMA structure for the errors.

The `xreg` argument was already used in the Data Report to pass the COVID dummy variable into both the headline and core models, which was the fix that allowed them to pass the Ljung-Box diagnostic test. Extending the model to include lagged gas prices would follow the exact same mechanism by adding a gas price column alongside the COVID dummy in the `xreg` matrix, with no other changes to the model structure.

A lagged gas price series from FRED (GASREGCOVW) would be the natural regressor here. Using a one-period lag means the value is already observed at the time of forecasting, which avoids the complication of needing to forecast the external variable before forecasting inflation.

```
# Illustrative ARIMAX -- not estimated in this report
gas <- fredr(series_id = "GASREGCOVW",
             observation_start = as.Date("2000-01-01"))

gas_lag1 <- lag(gas$value, 1)

arimax_model <- auto.arima(diff_log_headline, xreg = gas_lag1)
```

ARIMAX does not always beat a pure ARIMA benchmark. In cases where the external variable adds little predictive power beyond the series own history, AIC may not improve. For headline CPI during an acute energy shock, however, lagged oil prices carry genuinely useful information that the univariate model cannot replicate, making this a case where the extension would likely have reduced the forecast error.

4.4 Cointegration and the Error Correction Model

Another way to think about the oil and CPI relationship is through cointegration. If headline CPI and oil prices are both $I(1)$ and share a stable long-run relationship, they are said to be cointegrated. This can be tested using the Engle-Granger two-step procedure: first estimate the long-run regression of CPI on oil

prices, then run an ADF test on the residuals. If the residuals are stationary ($p < 0.05$), cointegration is confirmed and the two series move together over time even if they diverge in the short run.

When cointegration holds, the appropriate model is an Error Correction Model (ECM). The ECM captures both the short-run response to a shock and the longer-run process by which inflation gravitates back toward its equilibrium relationship with energy prices. The key parameter is the error correction coefficient δ , which must be negative for the system to be stable. A value of $\delta = -0.2$, for example, means 20% of any deviation from the long-run relationship is corrected each month. Simply differencing the data and running a standard ARIMA discards this long-run information entirely, which can leave the model underspecified when the two series are genuinely linked.

4.5 Structural Breaks and Model Stability

All time series models assume that the relationships they estimate remain stable over time. When a major economic event occurs, that assumption can break down. The Iran conflict caused a 21.2% monthly gasoline spike in March 2026, a shock that the model post-2000 was not trained on.

A rolling regression on the relationship between oil prices and CPI inflation would show the connection between the two series strengthening sharply around March 2026, as the energy shock made itself visible in the data. A Chow test would then formally check whether the coefficients before and after the conflict are statistically different from each other. If a break is confirmed, the standard fixes are to split the sample at that date and estimate separate models, add a dummy variable to flag the break period, or use a time-varying parameter model where the coefficients are allowed to shift gradually over time rather than being held constant throughout the entire sample.

4.6 VAR as a Multivariate Framework

A further extension is to model both oil prices and CPI inflation together using a Vector Autoregression (VAR). The key difference from ARIMAX is that ARIMAX only allows oil prices to affect CPI in one direction, while a VAR allows both variables to influence each other. Oil prices affect inflation, but inflation expectations can also feed back into commodity markets. In a VAR, both variables are treated as connected and each one is explained by the past values of both.

For this setup, oil prices would be listed first in the VAR since they are more externally driven by geopolitical events, and CPI inflation second since it responds to energy costs rather than driving them. Two tools make the VAR output useful in practice. An Impulse Response Function (IRF) shows how CPI inflation reacts over the following months after a sudden spike in oil prices, capturing both how large the effect is and how long it lasts before fading. Forecast Error Variance Decomposition (FEVD) answers a different question: out of all the uncertainty in a CPI forecast, how much of it comes from oil price movements versus CPI's own historical patterns. Together these two tools give a much clearer picture of how energy shocks feed through to inflation than a univariate ARIMA by itself.

5. Forecast Visualization

```
headline_raw <- fredr(series_id = "CPIAUCSL",
                      observation_start = as.Date("2000-01-01"))
core_raw      <- fredr(series_id = "CPILFESL",
                      observation_start = as.Date("2000-01-01"))
```

```

headline_clean <- headline_raw %>% arrange(date) %>% filter(!is.na(value))
core_clean      <- core_raw      %>% arrange(date) %>% filter(!is.na(value))

headline_data <- headline_clean %>%
  mutate(inflation_yoy = (value / lag(value, 12) - 1) * 100) %>%
  filter(!is.na(inflation_yoy))

log_headline_ts <- ts(log(headline_clean$value), start = c(2000, 1), frequency = 12)
log_core_ts     <- ts(log(core_clean$value),     start = c(2000, 1), frequency = 12)
diff_log_headline <- diff(log_headline_ts)
diff_log_core    <- diff(log_core_ts)

covid_dummy_h <- as.numeric(
  headline_clean$date[-1] >= as.Date("2021-03-01") &
  headline_clean$date[-1] <= as.Date("2022-12-01")
)
covid_dummy_c <- as.numeric(
  core_clean$date[-1] >= as.Date("2021-03-01") &
  core_clean$date[-1] <= as.Date("2022-12-01")
)

final_headline <- auto.arima(diff_log_headline,
                             xreg = matrix(covid_dummy_h, ncol = 1),
                             stepwise = FALSE, approximation = FALSE)
final_core     <- auto.arima(diff_log_core,
                             xreg = matrix(covid_dummy_c, ncol = 1),
                             stepwise = FALSE, approximation = FALSE)

future_dummy <- matrix(rep(0, 12), ncol = 1)
fc_headline <- forecast(final_headline, h = 12, xreg = future_dummy, level = c(50, 80))
fc_core     <- forecast(final_core,      h = 12, xreg = future_dummy, level = c(50, 80))

last12_log_h <- tail(log(headline_clean$value), 12)
last12_log_c <- tail(log(core_clean$value), 12)

convert_to_yoy <- function(fc_mean, last12_log) {
  last_log_val <- tail(last12_log, 1)
  log_forecast_levels <- last_log_val + cumsum(as.numeric(fc_mean))
  yoy <- (exp(log_forecast_levels) / exp(last12_log) - 1) * 100
  return(yoy)
}

yoy_headline_point <- convert_to_yoy(fc_headline$mean, last12_log_h)
yoy_headline_lo50 <- convert_to_yoy(fc_headline$lower[,1], last12_log_h)
yoy_headline_hi50 <- convert_to_yoy(fc_headline$upper[,1], last12_log_h)
yoy_headline_lo80 <- convert_to_yoy(fc_headline$lower[,2], last12_log_h)
yoy_headline_hi80 <- convert_to_yoy(fc_headline$upper[,2], last12_log_h)

forecast_dates <- seq(as.Date("2026-03-01"), by = "month", length.out = 12)

headline_fc_df <- data.frame(
  date = forecast_dates,
  point = yoy_headline_point,

```

```

lo50 = yoy_headline_lo50,
hi50 = yoy_headline_hi50,
lo80 = yoy_headline_lo80,
hi80 = yoy_headline_hi80
)

actual_point <- data.frame(
  date = as.Date("2026-03-01"),
  value = 3.3
)

march_forecast <- data.frame(
  date = as.Date("2026-03-01"),
  point = 2.68,
  lo50 = 2.50,
  hi50 = 2.85,
  lo80 = 2.35,
  hi80 = 3.01
)

march_actual <- data.frame(
  date = as.Date("2026-03-01"),
  value = 3.30
)

```

5.1 Full History

```

last_obs <- headline_data %>% filter(date == max(date))

ggplot() +

  geom_line(data = headline_data, aes(x = date, y = inflation_yoy),
    color = "#2c7bb6", linewidth = 0.5) +

  geom_segment(aes(x = last_obs$date,
    xend = as.Date("2026-03-01"),
    y = last_obs$inflation_yoy,
    yend = 2.68),
    color = "#2c7bb6", linewidth = 0.5, linetype = "dashed") +

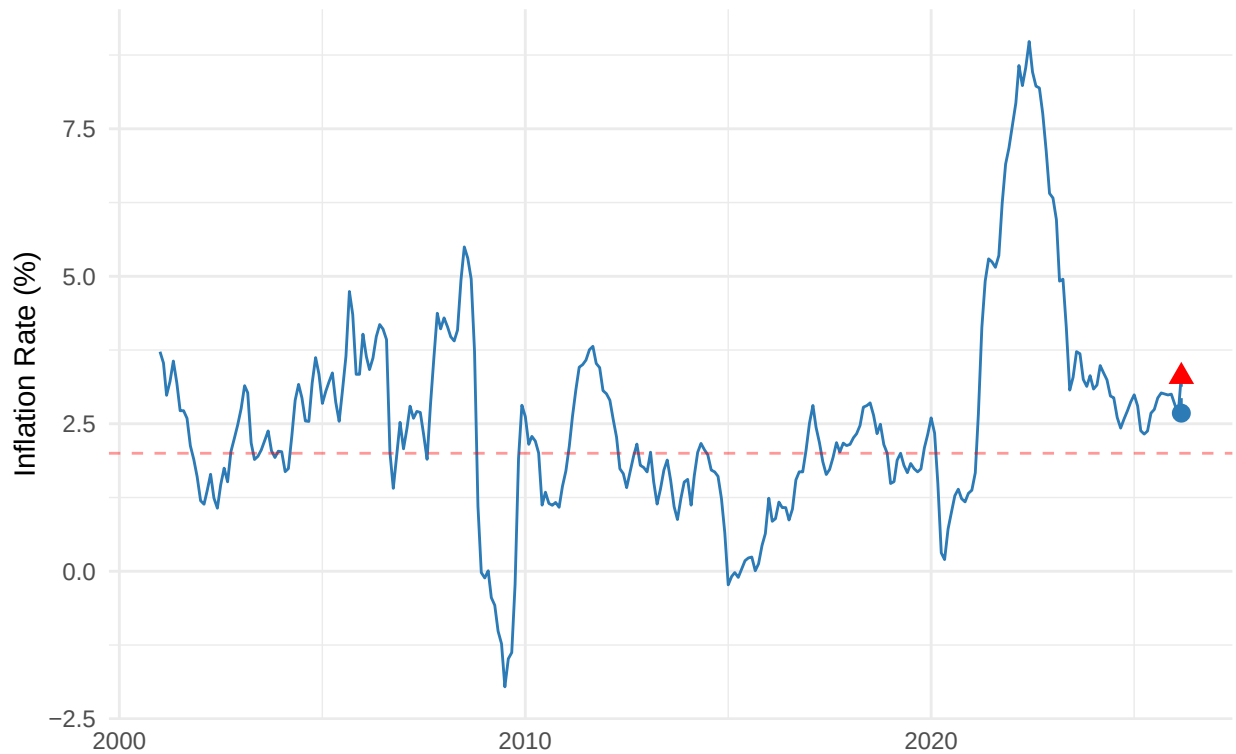
  geom_point(data = march_forecast, aes(x = date, y = point),
    color = "#2c7bb6", size = 3, shape = 16) +

  geom_point(data = march_actual, aes(x = date, y = value),
    color = "red", size = 3, shape = 17) +
  geom_hline(yintercept = 2, linetype = "dashed", color = "red", alpha = 0.4) +
  labs(title = "Headline CPI Inflation Forecast vs. Actual: March 2026",
    subtitle = "Full history (2001-2026) | Blue dot = forecast (2.68%) | Red triangle = actual (3.30%)",
    x = NULL, y = "Inflation Rate (%)") +
  theme_minimal(base_size = 11) +
  theme(plot.title = element_text(face = "bold"))

```

Headline CPI Inflation Forecast vs. Actual: March 2026

Full history (2001–2026) | Blue dot = forecast (2.68%) | Red triangle = actual (3.30%)



5.2 Zoomed View: March 2026 Forecast vs. Actual

```
headline_data_zoomed <- headline_data %>%  
  filter(date >= as.Date("2023-01-01") & date <= as.Date("2026-02-01"))  
  
ggplot() +  
  
  geom_line(data = headline_data_zoomed, aes(x = date, y = inflation_yoy),  
    color = "#2c7bb6", linewidth = 0.6) +  
  
  geom_segment(aes(x = as.Date("2026-02-01"),  
    xend = as.Date("2026-03-01"),  
    y = headline_data_zoomed$inflation_yoy[nrow(headline_data_zoomed)],  
    yend = 2.68),  
    color = "#2c7bb6", linewidth = 0.6, linetype = "dashed") +  
  
  geom_errorbar(data = march_forecast,  
    aes(x = date, ymin = lo80, ymax = hi80),  
    color = "#2c7bb6", width = 10, linewidth = 0.8, alpha = 0.5) +  
  
  geom_errorbar(data = march_forecast,  
    aes(x = date, ymin = lo50, ymax = hi50),
```



```

    color = "#2c7bb6", width = 10, linewidth = 1.4, alpha = 0.8) +

geom_point(data = march_forecast, aes(x = date, y = point),
  color = "#2c7bb6", size = 3.5, shape = 16) +

geom_point(data = march_actual, aes(x = date, y = value),
  color = "red", size = 3.5, shape = 17) +

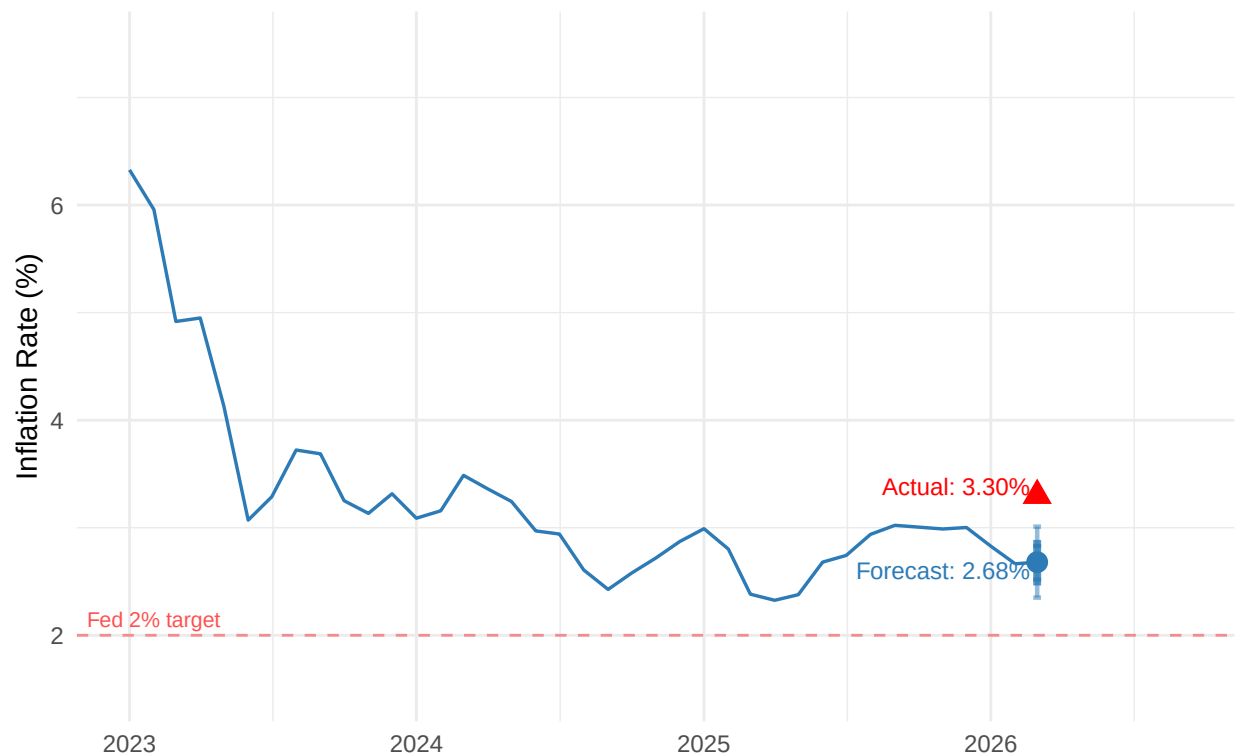
annotate("text", x = as.Date("2026-02-20"), y = march_forecast$point - 0.08,
  label = "Forecast: 2.68%", color = "#2c7bb6", size = 3.0, hjust = 1) +
annotate("text", x = as.Date("2026-02-20"), y = march_actual$value + 0.08,
  label = "Actual: 3.30%", color = "red", size = 3.0, hjust = 1) +

geom_hline(yintercept = 2, linetype = "dashed", color = "red", alpha = 0.4) +
annotate("text", x = as.Date("2023-02-01"), y = 2.15,
  label = "Fed 2% target", color = "red", size = 2.8, alpha = 0.7) +
coord_cartesian(xlim = c(as.Date("2023-01-01"), as.Date("2026-09-01")),
  ylim = c(1.5, 7.5)) +
labs(title = "Headline CPI Forecast vs. Actual: March 2026",
  subtitle = "Blue dot = point forecast (2.68%) | Thick bar = 50% interval | Thin bar = 80% interval",
  x = NULL, y = "Inflation Rate (%)") +
theme_minimal(base_size = 11) +
theme(plot.title = element_text(face = "bold"))

```

Headline CPI Forecast vs. Actual: March 2026

Blue dot = point forecast (2.68%) | Thick bar = 50% interval | Thin bar = 80% interval | Red



The full history chart provides context across the entire post-2000 sample. The zoomed view focuses on the evaluation. The blue dot is the submitted point forecast of 2.68%, the thick bar shows the 50% interval

[2.50, 2.85], and the thin bar shows the 80% interval [2.35, 3.01]. The red triangle is the actual BLS release of 3.30%, sitting just above the upper bound of the 80% interval. The miss was an extreme outcome driven by an unusual shock rather than a sign the model was consistently off in one direction.

6. Discussion

The core result of this evaluation is straightforward. The ARIMA model produced an accurate forecast for core CPI (2.72% vs. actual 2.60%) and missed headline CPI by 0.62 percentage points. The entire headline miss is explained by the energy component: gasoline prices rose 21.2% in March 2026 following the Iran conflict, an event that occurred after the forecast was submitted and that no backward-looking time series model could have anticipated.

This outcome is consistent with what the Data Report's limitations section predicted before the release. Identifying in advance that the model would underforecast headline CPI due to an energy shock, and then observing the miss in exactly that direction, demonstrates that the model's limitations were well understood even if they could not be avoided given the available data at the time of submission.

The miss also illustrates why testing for structural breaks matters in practice. A rolling regression on the inflation-energy relationship would show the pass-through coefficient shifting sharply around March 2026, as geopolitical supply disruption changed the typical relationship between energy costs and consumer prices. ARIMA assumes that the patterns it learns from historical data stay consistent over time. A sudden supply shock like the Iran conflict is exactly the kind of event that violates that assumption, since it changes the relationship between energy prices and inflation in a way the model has no way to anticipate.

Looking forward, two extensions would address the headline miss most directly. An ARIMAX model with lagged gas prices as an external regressor would allow the ARIMA to partially incorporate energy price information, and since a one-period lag is used the gas price value is already observed at the time of forecasting so no additional forecast is needed. A bivariate VAR treating both oil prices and CPI as jointly evolving would go further, allowing feedback in both directions and enabling the use of Impulse Response Functions to quantify the typical size and duration of the oil price pass-through into inflation.

7. Conclusion

```
conclusion <- data.frame(
  Item = c(
    "Headline forecast (submitted)",
    "Headline actual (April 10)",
    "Headline error",
    "Headline 80% interval coverage",
    "Core forecast (submitted)",
    "Core actual (April 10)",
    "Core error",
    "Core 80% interval coverage"
  ),
  Value = c(
    "2.68%",
    "3.30%",
```

```

    "+0.62 pp (underforecast)",
    "No: actual just above upper bound [2.35, 3.01]",
    "2.72%",
    "2.60%",
    "-0.12 pp (overforecast)",
    "Yes: actual within [2.59, 2.85]"
  )
)

kable(conclusion,
      col.names = c("Item", "Result"),
      align = "lr",
      caption = "Final Evaluation Summary")

```

Table 3: Final Evaluation Summary

Item	Result
Headline forecast (submitted)	2.68%
Headline actual (April 10)	3.30%
Headline error	+0.62 pp (underforecast)
Headline 80% interval coverage	No: actual just above upper bound [2.35, 3.01]
Core forecast (submitted)	2.72%
Core actual (April 10)	2.60%
Core error	-0.12 pp (overforecast)
Core 80% interval coverage	Yes: actual within [2.59, 2.85]

The ARIMA model produced one accurate forecast and one meaningful miss. The core CPI forecast was nearly exact, confirming the model correctly captured the underlying inflation trend. The headline miss was caused by the Iran conflict driving a sharp energy price spike that arrived after the forecast was submitted and was correctly anticipated as a model limitation in the Data Report.

The broader takeaway is that ARIMA is a reliable benchmark for forecasting inflation in stable conditions but cannot respond to sudden exogenous shocks. Incorporating energy prices through ARIMAX or a full VAR system would make the model more robust in volatile periods.